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RESEARCH ARTICLE

Content Quality Over Functionality: How Aesthetics, Authenticity, and Creativity Drive Designers' Adoption of AI-Generated Video Advertising Tools

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ABSTRACT This study examines how perceived content quality in AI-generated video advertisements influences advertising designers' intention to adopt AI-generated video advertising tools. Extending the Technology Acceptance Model (TAM), it models aesthetics, authenticity, and creativity as antecedents of perceived usefulness (PU) and perceived trust (PT), which in turn affect behavioral intention. A stimulus-based survey design was used. Six mainstream AI video-generation tools were applied under a unified prompt and a shared perfume-advertising task, and the screened outputs were compiled into a standardized 33-second stimulus video. After viewing the same video, 556 valid respondents completed a structured questionnaire, and the data were analyzed using partial least squares structural equation modeling (PLS-SEM). The results show that aesthetics, authenticity, and creativity are all associated with adoption intention, but through different mechanisms. Authenticity and creativity show more stable indirect effects through perceived usefulness and perceived trust, whereas aesthetics mainly exerts a direct effect. Creativity has the strongest effects on perceived usefulness and perceived trust. PU also positively predicts PT, underscoring the importance of trust in professional creative settings characterized by uncertain AI outputs. These findings offer a content-centered extension of TAM, showing that perceived content quality is associated with designers' adoption intention toward AI-generated advertising tools, with implications for AI tool development, human–AI advertising workflows, and AI literacy in creative education.

INDEX TERMS AI-generated video advertising, technology acceptance model (TAM), AI video tools, behavioral intention, perceived trust, perceived usefulness.

I. INTRODUCTION

The rapid development of Artificial Intelligence (AI) has radically changed the way advertising is created [1]. AI-generated content (AIGC) is a major asset for advertisers, offering substantial advantages over traditional manual workflows [2]. In particular, AI technologies can produce high-quality visual material at remarkable speed, streamline the creative process, and meet the industry's increasingly demanding standards for aesthetics, authenticity, and innovation [3], [4]. AI is widely recognized as one of the most transformative forces driving emerging technological fields. Leveraging advances in deep learning, computer vision, and natural language processing, AI systems can autonomously generate and optimize advertising copy, images, and videos in alignment with brand identity, thereby enhancing creative efficiency and significantly reducing production costs [5]. Recent scholarship has further shown that AI is reshaping advertising through targeting, personalization, content creation, and ad optimization, while also introducing new ethical and managerial challenges [6].

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At present, leading video generation platforms such as Runway, Luma AI, Kling AI, Ji Meng AI, Hai Luo AI, and Vidu AI are increasingly being integrated into advertising workflows, not only to improve production efficiency but also to support creative ideation and content generation. This shift is consistent with prior research suggesting that AI is increasingly shaping advertising creativity rather than merely supporting execution [7]. As an important sub-field of AIGC, AI-generated advertising can extract, reconstruct, and synthesize information on creative conception, editing, and film production [8], [9], thereby improving the visibility and overall impact of advertising. However, with the rapid popularity of generative AI (Gen AI) on the consumer and enterprise sides, concerns about content quality, consistency and credibility are also growing among designers [10]. Despite the significant advantages of AI tools, designers' willingness to adopt them still varies; however, many practitioners recognize the effective value of AI tools for creative processes [11].

Most existing studies examine the impact of AI-generated advertisements from the consumers' perspective, focusing on their role in consumer behavior, brand attitude and purchase intention [12]. Research on how advertising designers adopt AI tools is still relatively limited. Designers' adoption decisions may be affected by multiple factors, including visual output quality, content authenticity, and creative uniqueness [13]. In addition, TAM has been widely used to explain users' technology adoption behavior [14]; however, there is still a lack of systematic research on how the specific attributes of AI-generated content such as aesthetic attraction, authenticity, and creativity affect the behavioral intention of designers through the mediating roles of perceived usefulness and perceived trust.

Accordingly, this article focuses on the following core research issues: 1. What content quality attributes (aesthetics, authenticity, and creativity) of video advertisements generated by artificial intelligence will significantly affect the designer's intention to adopt artificial intelligence video tools? 2. How do perceived usefulness and perceived trust mediate the relationship between content perceptions and behavioral intention? 3. In the context of generative artificial intelligence with uncertain outcomes, how do perceived usefulness and perceived trust interact to influence designers' behavioral intention?

To address these research questions, this study makes three important contributions to the existing literature. First, focusing on the professional creative use of generative AI, this study provides a contextualized and content-centered extension of the Technology Acceptance Model (TAM) by introducing aesthetics, authenticity, and creativity as core content-quality antecedents of advertising designers' adoption of AI-generated video advertising tools. Second, this study incorporates perceived trust alongside perceived usefulness to better explain adoption decisions in generative-AI contexts characterized by high output uncertainty. Third, it provides empirical evidence from advertising designers, thereby enriching current research on the application of generative AI and human–AI collaboration in creative industries.

The research framework of this article is as follows: Section II surveys prior scholarship on advertising and TAM, outlining the TAM architecture, the attributes of AI-produced content and its use within advertising, and advancing the study's hypotheses and conceptual framework. Section III details the research methods and experimental design. Section IV describes data collection and examines the sample. Section V presents empirical findings and tests of hypotheses. Section VI offers a discussion, contrasts with earlier studies, and clarifies the theoretical and practical implications. Section VII delineates study limitations and future avenues. Section VIII concludes, providing insights that support the adoption of AI tools in advertising.

II. LITERATURE REVIEW AND HYPOTHESES

A. TECHNOLOGY ACCEPTANCE MODEL (TAM)

The Technology Acceptance Model (TAM), proposed by Davis, has been widely used to explain technology adoption [15]. In this study, perceived usefulness (PU) remains central because it reflects whether designers believe AI-generated video advertising tools can support creative tasks and improve professional performance. However, generative-AI advertising tools differ from conventional technologies in that they not only assist execution but also produce visual and narrative outputs that may directly enter the final advertisement [16]. Therefore, designers' adoption judgments depend not only on utility but also on whether the generated outputs are credible and professionally reliable.

For this reason, the present study incorporates perceived trust (PT) alongside PU to explain behavioral intention. Perceived ease of use (PEOU) is not emphasized as a focal mechanism because, in professional advertising contexts, designers are likely to value the usefulness, trustworthiness, and creative suitability of AI-generated outputs more than interface simplicity alone. Accordingly, this study extends TAM as a content-centered framework in which perceptions of aesthetics, authenticity, and creativity influence behavioral intention through perceived usefulness and perceived trust.

B. CONTENT QUALITY OF AI-GENERATED ADVERTISING

AI-generated content (AIGC) is a key subfield of AI [17]. Relying on deep learning and computer vision, AIGC can independently generate text, images, audio, and video [18]. AIGC has seen broad uptake in art design and video production, markedly boosting production efficiency. Mainstream video generation tools such as Runway, Luma AI, Kling AI, Ji Meng AI, Hai Luo AI, and Vidu AI have made significant progress in automatic editing, intelligent synthesis, and special-effects generation, thereby improving the efficiency and quality of video production.

In video advertising, AI tools are increasingly used to support creative efficiency, content production, and audience interaction [19]. AI tools are also reshaping the advertising production process. However, designers are still concerned about the role of AI in creative conception, visual presentation, and content quality [18] (Figure 1).

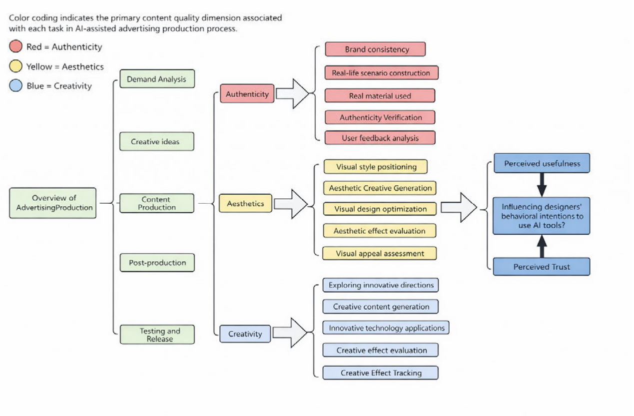

FIGURE 1. Analysis of the impact of AI advertising production process on designer behavior intention based on TAM.

The uniqueness of AI-generated advertising is that, unlike traditional design software that only executes human instructions, AI systems exhibit autonomous generation and probabilistic output, actively participate in the creative process, and introduce algorithmic creativity, randomness, and opacity. As a result, designers face new challenges such as reduced creative controllability, uncertainty in output quality, and vague originality. Recent studies have also found differences in how content with "AI logos" is perceived by audiences and creators: for example, trust and credibility in content bearing AI logos are lower, and perceptions of the dynamism and emotional resonance of AI-generated artworks are reduced [20]. In addition, AI-generated advertisements for luxury brands reduce audiences' perceptions of the fit between investment effort and brand advertising [21]. This suggests that, even when efficiency is improved, AI-generated content may trigger doubt and weaken perceived authenticity.

Therefore, compared with conventional technologies, the adoption of AI in advertising depends not only on function or ease of use, but also on trust, perceived authenticity, and creative controllability. Overall, the content quality of AI-generated advertisements is mainly reflected in three dimensions: authenticity, aesthetics, and creativity. Aesthetics refers to the visual components and artistic quality of an advertisement, which have a great influence on designers' emotional reactions and the overall appeal of the advertisement [22]. Authenticity is related to the consistency between advertisement content and brand identity; therefore, AI outputs must be credible enough to guarantee effectiveness [23]. Creativity reflects the innovation and originality of advertisement content, which is essential for encouraging designers and improving advertising quality [24], [25]. These qualities not only influence advertising communication effects but may also shape designers' willingness to use AI-based advertising tools.

In the production process, authenticity verification, aesthetic enhancement, and creative generation are key links in the quality control of AI advertising content. In summary, AI video tools are promoting the evolution of the advertising industry toward a more efficient, intelligent, and personalized creative mode [2], and authenticity, aesthetics, and creativity remain the core factors affecting advertising quality.

C. HYPOTHESES DEVELOPMENT

1) AESTHETICS

The aesthetics of advertising refers to its visual design and creative attractiveness, which can affect designers' emotional experience and the persuasiveness of the advertisement itself [26]. This includes elements such as color, layout, imagery, and composition, which together constitute the advertising style and influence the audience's psychological responses [27]. In the context of AI-generated advertising, aesthetics determines the visual quality of the content, which may affect designers' perceptions and acceptance of AI tools.

In the present study, respondents evaluated AI-generated perfume advertisements rather than abstract system interfaces or generic media outputs. Such advertising tasks rely heavily on visual atmosphere, product salience, stylistic coherence, and emotional tone. Therefore, in this task setting, aesthetic quality functions not merely as a surface feature but as a practical cue indicating whether the AI tool can support professional advertising production and creative performance.

Aesthetics can significantly affect perceived usefulness. Visually attractive designs improve designers' evaluations of the usability and value of AI tools [28]. Highly appealing advertising outputs can also stimulate creativity and help designers develop conceptual ideas more efficiently [29]. In AI-generated video advertising scenarios, especially in visually driven tasks such as perfume advertising, higher aesthetic quality may lead designers to perceive AI tools as more capable of supporting both production efficiency and creative performance, thereby enhancing perceived usefulness. This is because aesthetically refined outputs are more likely to reduce the need for repeated visual correction and make the generated results appear more usable in real advertising practice.

Aesthetics is also pivotal in forming perceived trust. Beyond visual appeal, aesthetically refined outputs can directly strengthen designers' trust in AI tools by creating an immediate impression of reliability, and this effect may persist during continued use. Aesthetics can therefore influence trust progressively: careful and coherent visual design not only shapes first impressions but also reinforces longer-term confidence in the tool [30]. In addition, consistent aesthetic judgments throughout advertising design can improve perceived credibility and audience confidence. Given that visual output is central to AI-generated video advertising, aesthetic excellence may become an important basis for building and maintaining designers' trust, because polished and stylistically coherent outputs are more likely to be interpreted as signals of quality control, technical competence, and professional reliability.

Aesthetic features may also significantly enhance designers' behavioral intention (BI). High-quality advertising visuals can stimulate designers' emotional resonance and creative interest, and increase their motivation to use AI tools [27]. Such aesthetic appeal can further improve designers' evaluations of AI tools and strengthen their intention to use them [31]. In advertising design, where visual impact is central to both concept development and presentation, tools that consistently generate aesthetically compelling outputs are more likely to be adopted in future creative work. Accordingly, the effect of aesthetics in this study should be understood within a content-evaluation task in advertising design rather than as a purely interface-level perception of technology.

H1: The aesthetics of AI-generated video advertisements positively influence designers' perceived usefulness.

H2: The aesthetics of AI-generated video advertisements positively influence designers' perceived trust.

H3: The aesthetics of AI-generated video advertisements positively influence designers' behavioral intention.

2) AUTHENTICITY

In contemporary advertising discourse, authenticity is conceptualized as a multidimensional construct encompassing realism, honesty, and consistency with brand identity and tone [32]. It refers to the extent to which content truthfully conveys brand value, ensures sincere alignment between the creative output and the core brand identity, and evokes emotional resonance through transparent and non-deceptive representation. Consequently, authenticity serves as a key criterion for evaluating advertising quality and plays an important role in strengthening designers' confidence while reducing uncertainty during the creative process [33]. In the specific context of AI-generated video advertising, these dimensions are manifested through the realism of scene construction, the credibility of the materials used, and the consistency of narrative and brand expression [34], [35].

Authenticity can significantly influence perceived usefulness. Within the TAM framework, authenticity is closely related to whether designers regard AI-generated outputs as practically valuable for real advertising tasks. When AI-generated advertisements show a high level of authenticity and fit well with brand identity and market expectations, designers are more likely to perceive AI tools as useful for advertising creation [36]. For example, AI can achieve realistic visual effects through credible materials and visual enhancement, thereby reducing the need for manual adjustment and improving production efficiency [37]. In professional advertising work, authenticity also helps generated outputs appear more ready for use in brand communication, which further strengthens designers' perceptions of the tool's usefulness.

Authenticity can also significantly influence perceived trust. In AI advertising creation, assessing authenticity requires designers to evaluate the materials, scene construction, and narrative logic generated by AI in order to determine whether the outputs meet professional expectations [38]. Advertisements that achieve a higher level of authenticity can not only strengthen the persuasiveness of content, but also improve designers' trust in AI-generated results [39], [40]. This is because authenticity signals that the generated content is not merely visually convincing, but also sufficiently credible and appropriate for actual advertising communication. Such trust can improve designers' satisfaction with the tools and may promote their wider use in real advertising practice.

Authenticity may also positively influence designers' behavioral intention (BI). When advertising content is close to reality and effectively conveys brand value [41], designers are more inclined to adopt advanced tools and integrate them into the creative process. Realistic and authentic advertising content can trigger emotional resonance and enhance designers' creative engagement, thereby strengthening their motivation to use AI tools [23]. In addition, when AI-generated outputs present convincing realism and narrative consistency, designers may regard these tools as more professionally dependable and more suitable for future projects, which directly increases their intention to adopt them.

H4: The authenticity of AI-generated video advertisements positively influences designers' perceived usefulness.

H5: The authenticity of AI-generated video advertisements positively influences designers' perceived trust.

H6: The authenticity of AI-generated video advertisements positively influences designers' behavioral intention.

3) CREATIVITY

Creativity in advertising is widely recognized as a dual construct encompassing novelty and usefulness. It refers to the extent to which advertising content is perceived as highly original and unexpected, which represents novelty, while also being practically valuable and strategically appropriate for solving communication problems, which defines its usefulness [25]. Creative advertising enables messages to stand out from a large amount of competing information, effectively attract audience attention, and improve communication efficiency [42]. In the context of AI-generated video advertising, creativity requires the AI to go beyond merely generating visually unique anomalies. Instead, it must demonstrate novelty through distinctive concepts and original expressive forms, while proving its usefulness by successfully engaging the target audience and fulfilling professional advertising objectives.

Innovative advertising can achieve two important goals: it attracts audience attention and also triggers emotional investment, thereby improving brand recognition and purchase intention. From the designer's perspective, such creative potential may also strengthen perceived usefulness. When

AI-generated advertisements present novel and meaningful creative ideas, designers are more likely to regard AI tools as useful resources for idea exploration, concept expansion, and early-stage creative development. In advertising production, creativity is valuable not simply because it makes content more attractive, but because it helps designers generate alternative directions, accelerate concept selection, and broaden the range of workable ideas. Therefore, stronger creativity in AI-generated advertising is expected to enhance designers' perceived usefulness of AI tools [43].

In addition to increasing the attractiveness of advertising, creativity may also shape designers' perceived trust. AI-generated advertising relies on algorithmic models to extract, combine, and reorganize information from large-scale data sources, thereby demonstrating strong innovative potential [44]. When the outputs generated by AI are not only novel but also meaningful and relevant to advertising goals, designers may infer that the tool is capable of producing valuable and professionally applicable results. In this sense, creativity can strengthen trust not merely because it signals innovation, but because it suggests that the tool can consistently support high-level creative work. As a result, highly creative outputs may increase designers' confidence in the practical value and reliability of AI tools.

Creativity can also effectively increase behavioral intention (BI). Creative advertising enhances not only content attractiveness but also designers' interest in and reliance on the tools used to produce it. AI-generated video advertisements are particularly notable in content personalization and may exhibit greater uniqueness in visual presentation and conceptual design, thereby improving creativity [45]. For professional designers, the acceptability of advertising tools often depends on whether they combine innovation with practical usability [6]. Therefore, when AI-generated advertising outputs demonstrate strong creativity, designers are more likely to perceive the tools as worth integrating into future advertising production, which in turn strengthens their behavioral intention to continue using them.

H7: The creativity of AI-generated video advertisements positively influences designers' perceived usefulness of AI tools.

H8: The creativity of AI-generated video advertisements positively influences designers' perceived trust in AI tools.

H9: The creativity of AI-generated video advertisements positively influences designers' behavioral intention.

4) PERCEIVED USEFULNESS

In the AIGC-based advertising design environment, perceived usefulness reflects designers' recognition of the effectiveness of AI tools in improving creative efficiency, enhancing advertising quality, and stimulating innovative ideas [46]. In the present study, perceived usefulness specifically refers to whether designers believe that AI-generated video advertising tools can provide meaningful support for real creative tasks and professional advertising production.

Perceived usefulness plays a key role in shaping behavioral intention (BI). Existing studies have shown that PU directly affects users' willingness to adopt technology and thereby influences their behavioral choices [47]. When designers perceive that AI tools can effectively support their creative tasks by improving efficiency, reducing repetitive work, and providing novel solutions, their acceptance of the technology is likely to increase significantly. In the context of AI-generated advertising, if designers regard such tools as practically valuable for concept development, visual production, and creative decision-making, they will be more inclined to adopt them and show a stronger intention to continue using them. Therefore, higher perceived usefulness is expected to directly strengthen designers' behavioral intention.

Perceived usefulness can also positively influence perceived trust. The impact of perceived usefulness on perceived trust has been widely recognized because users often interpret strong task performance as an indication of system reliability and credibility [48]. In AI-generated advertising, when designers perceive that AI tools can continuously produce high-quality content consistent with brand tone and design standards, their trust in the technology is likely to increase. This is because usefulness in this context is not only associated with efficiency, but also with the ability of the tool to deliver outputs that are professionally relevant, stable, and dependable for real advertising work. When designers repeatedly perceive practical value in the generated results, they are more likely to infer that the tool is reliable enough to be trusted in future creative projects.

H10: Perceived usefulness positively influences designers' behavioral intention.

H11: Perceived usefulness positively influences designers' perceived trust.

5) PERCEIVED TRUST

Perceived trust refers to users' subjective evaluation of an AI tool in terms of reliability, consistency, and credibility. In AI-mediated creative work, trust functions as an important mechanism for reducing uncertainty about output quality, professional appropriateness, and potential reputational consequences [49]. In the context of AI-generated video advertising, designers are unlikely to adopt such tools if they doubt whether the generated outputs can consistently meet brand tone, visual coherence, and practical production standards. Therefore, perceived trust is expected to directly strengthen designers' behavioral intention by increasing their confidence that AI tools can be integrated into real advertising workflows with acceptable professional reliability [50]. Prior studies have also demonstrated that trust plays a crucial role in technology adoption decisions, particularly in contexts characterized by uncertainty and perceived risk [9]. Accordingly, the following hypothesis is proposed. (Figure 2).

H12: Perceived trust positively influences designers' behavioral intention (BI).

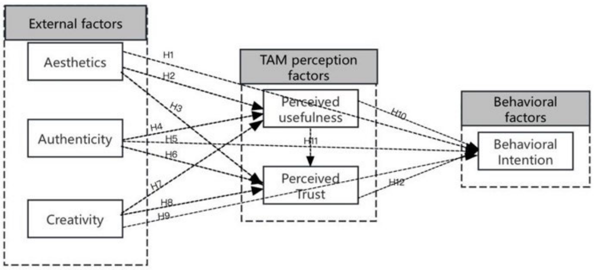

FIGURE 2. Proposed conceptual model.

III. RESEARCH METHODS

This study adopted a stimulus-based survey design to investigate how advertising designers assess the content quality of AI-generated video advertisements and how these assessments shape perceived usefulness, perceived trust, and adoption intention. The research consisted of three stages. First, six mainstream AI video-generation tools were used to generate perfume advertisement videos under a unified prompt. Second, the generated outputs were reviewed and screened by experts to establish the final stimulus material. Third, designers were invited to view the standardized AI-generated video advertisement and complete a structured questionnaire, enabling a quantitative analysis of their perceptions and adoption intentions. The following subsections detail the AI tool selection and generation process, the stimulus and questionnaire design, and the sample recruitment and survey implementation procedures.

A. SELECTION OF AI TOOLS AND EXPERIMENTAL DESIGN

To ensure that the experimental materials accurately reflect the current technical level, this study first conducted an exploratory review of the mainstream AI video generation tools widely used in China's creative industry. By collecting popularity data on multiple social platforms such as Bilibili, Red Book, Zhihu and Weibo, the research team analyzed the frequency of keywords related to "AI video generation" and "text-to-video". Based on user activity, content volume and platform mention frequency, six most popular tools were selected: Runway, Luma AI, Hailuo, Ji Meng, Kling and Vidu (Table 1). These tools represent the mainstream technical path in generative video production, such as the synthesis of text-generated video (T2V) and image-generated video (I2V).

To ensure that the AI-generated advertising videos used as visual stimuli met professional quality standards and industry expectations, this study invited three experienced creative practitioners to participate in the stimulus development stage, including an advertising designer, a creative director, and a video editor. They were involved in the generation and evaluation of the AI-produced advertising clips, including prompt refinement, style adjustment, output review, and material screening. Their professional expertise helped improve the standardization, industry relevance, and visual quality of the final stimulus materials. Their background information is presented in Table 1. (These three professionals participated only in the stimulus development and screening stage and were not included in the questionnaire survey sample.)

TABLE 1. Professionals involved in the generation and evaluation of the stimulus video advertisement.

Designer	Gender	Work Experience	Professional Role
A1	Male	8 years	Advertising Designer
A2	Male	6 years	Creative Director
A3	Female	7 years	Video Editor

Figure 3 illustrates an AI-assisted workflow for video advertisement design. First, designers define the advertising goal, target audience, and brand requirements, and then use ChatGPT to translate these initial design requirements into structured prompts and draft advertising scripts. Next, Midjourney is employed to generate key visual frames and storyboard images based on these prompts, providing visual references for subsequent production. These scripts and images are then imported into AI video generation tools (e.g., Runway, Vidu, Kling, Dream Machine) to synthesize short advertising videos. In the final stage, designers review, edit, and composite the AI-generated footage, adjust timing and visual details, and produce the finalized commercial. This process shows how AI tools function as auxiliary modules embedded in the professional advertising workflow rather than replacing human creative control.

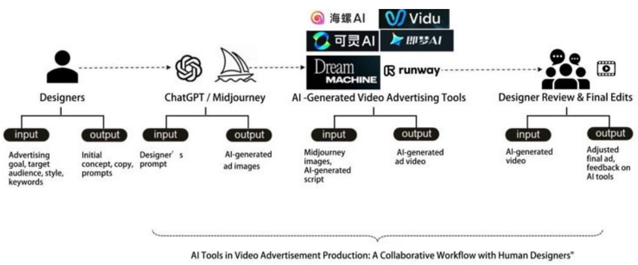

FIGURE 3. Collaborative mechanism of AI tools in the video advertising production process.

Based on this three-stage design, the following subsection details the selection of AI video-generation tools, the generation of stimulus materials, and the screening procedure.

This study selected perfume advertising as the stimulus theme for two main reasons. First, perfume advertisements typically rely on strong visual aesthetics, atmospheric storytelling, and abstract creative expression to convey brand value, which aligns closely with the three core constructs examined in this study: aesthetics, authenticity, and creativity. Second, perfume products evoke rich emotional and sensory associations, making them particularly suitable for evaluating the perceived quality of AI-generated video advertisements. To ensure consistency in theme, scene setting, and narrative structure across the six AI video-generation tools, the same prompt was applied to all tools. The prompt described a perfume advertising scene set in a misty forest at dawn, incorporating natural lighting, camera movement, product close-ups, and atmospheric visual expression. The full prompt was as follows: "At dawn, a tranquil forest bathed in golden light as sunlight filters through tall trees and mist. The camera glides over a shimmering river, capturing its gentle flow and reflections. The shot transitions to a close-up of a perfume bottle resting on moss-covered rocks, surrounded by dewy leaves and fog. The camera smoothly circles the square glass bottle containing pale blue liquid. As sunlight passes through the bottle, light particles float in the air, creating a warm and luxurious glow as the focus gradually tightens." It should be clarified that the seed strategy, generation randomness parameters, and underlying model sampling hidden parameters of the six AI tools used in this study were not available for user-defined adjustment or backtracking in their public versions, and thus were not recorded in our experiment.

During the stimulus development stage, the same three professionals reviewed the generated outputs and screened the clips. They evaluated the generated content according to four criteria: visual quality, motion smoothness, stylistic coherence, and advertising appropriateness. Specifically, visual quality referred to image clarity, lighting consistency, image detail, and the visibility of key objects; motion smoothness referred to the continuity and naturalness of camera movement and scene transitions; stylistic coherence referred to consistency with the unified prompt and the intended perfume-advertising tone; and advertising appropriateness referred to whether a clip could plausibly function as part of a perfume advertisement in terms of product salience, atmosphere, and brand-like presentation. Candidate clips that clearly failed to meet these baseline requirements were excluded. The screening process was criterion-based rather than based on a weighted numerical scoring scheme, with clips retained only when they satisfied the baseline professional requirements across these four dimensions. On this basis, the research team selected 12 representative clips from 38 candidate AI-generated materials and compiled them into one final stimulus video. The detailed screening criteria are provided in Appendix A.

Each clip lasted approximately 4–5 seconds, and the total duration of the compiled video was about 33 seconds. This relatively short duration was intended to reduce viewing fatigue while allowing respondents to observe multiple AI-generated styles under standardized exposure conditions. Figure 4 presents representative screenshots from the AI-generated advertising videos.

In this experiment, we systematically recorded and compared the performance of six AI video generation tools in the context of perfume advertising production. The evaluation dimensions include: tool models, key features, generated content characteristics, number of generations and generated parameters, which are used to evaluate the applicability of each tool in advertising creation (Table 2). We pay special attention to the performance of various tools in terms of visual quality, motion dynamics, aesthetic expression, authenticity and creative presentation. The multi-tool design treats AI tool variation as an experimental factor, while task type is controlled via a unified prompt, allowing the hypotheses to be tested in a diverse yet standardized context. The research objective is to explore the potential value and limitations of AI video generation technology in the advertising industry.

FIGURE 4. Video advertisement samples generated using six AI video tools.

TABLE 2. Specifications and performance comparison of AI tools used in the experiment.

AI Tool Name	Model	Key Features	Generated Content Characteristics	No. of Gens.	Generation Parameters
Runway	Gen-4 Alpha Turbo	Supports complex scene construction, intelligent frame interpolation	Rich scene details, natural dynamic expressions	4	1080×1920, Video Ratio: 9:16, Frame Rate: 24fps.
Luma AI Dream Machine	Video Model Ray 3	Intelligent analysis of key prompts, high clarity	Aesthetic scenes, detailed imagery	3	2560×1440, Video Ratio: 9:16, Frame Rate: 24fps.
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B. QUESTIONNAIRE DESIGN AND MEASUREMENT

After watching the advertising video generated by AI, the participants completed a structured online questionnaire to measure the six constructs in the extended Technology Acceptance Model (TAM): aesthetics (Ae), authenticity (Au), creativity (Cr), perceived usefulness, perceived trust and behavioral intention (BI).

The 30 measurement items were adapted from validated scales in prior studies, with the corresponding sources reported in Appendix B. All items were rated on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree). The questionnaire was originally written in English, and then translated into Chinese by three professional translators. The translation was reviewed by two advertising scholars and three creative industry practitioners to ensure terminological accuracy and contextual appropriateness.

Before the formal survey, a pre-test with 15 participants was conducted to evaluate the clarity, comprehensibility, and completion time of the questionnaire. According to the feedback, minor wording adjustments were made to improve the clarity and fluency of the questionnaire items.

C. PARTICIPANT RECRUITMENT AND PROCEDURE

The questionnaire was distributed online via the Questionnaire Star platform. A total of 568 participants were recruited through WeChat, QQ, Red Book and Weibo. The target group was advertising designers and creative practitioners working in major cities in China (such as Beijing, Shanghai, Guangzhou and Shenzhen). Before completing the questionnaire, all participants accessed the same online link and watched the same compiled 33-second AI-generated perfume advertisement. All clips were presented in a fixed sequence for all respondents. Immediately after viewing, participants completed the structured questionnaire so that their responses were based on a shared and recent visual exposure.

To improve participation and data quality, small monetary incentives (digital red envelopes) were provided to participants after they completed the questionnaire. Participation was completely voluntary and anonymous, and all respondents signed informed consent before participating. The study followed institutional ethical standards, and all data were used for academic purposes only.

IV. DATA COLLECTION AND SAMPLE CHARACTERISTICS

A. DATA COLLECTION AND SAMPLE DISTRIBUTION

We received 568 questionnaires; after data cleansing, 12 ineligible cases were removed, leaving 556 valid responses. The sample's profile appears in Table 3. Male respondents accounted for 51.6% (n = 287) and women 48.4% (n = 269). Most of the respondents were between 25–34 years old (36.5%, n = 203), followed by 35–44 years old (28.8%, n = 160), while the proportion of 18–24 years old (16.7%) and 45 years old and above (18.0%) was relatively low.

The sample consisted primarily of designers aged 25–44 years. Typically, this group may be more receptive to emerging technologies and tends to incorporate AI tools into their creative workflow. The low percentage of people over 45 years old may reflect a relatively lower reliance on emerging technologies.

TABLE 3. The description of characteristics in the sample.

Demographic	Frequency	%
Gender		
Male	287	51.6
Female	269	48.4
Age (years)		
18-24	93	16.7
25-34	203	36.5
35-44	160	28.8
45 years and above	100	18
Education level		
Below undergraduate	85	15.3
Undergraduate	293	52.7
Master	115	20.7
PhD	63	11.3
Field of work or study		
Visual Design	84	15.1
Advertising creative design	36	6.5
Video Advertising Design	90	16.2
Graphic Advertising Design	17	3.1
Animation Design	44	7.9
Multimedia Design	47	8.5
Network Advertising Design	54	9.7
Advertising copywriting planning	18	3.2
Advertising Photography	27	4.9
Brand planning and design	18	3.2
Interactive advertising design	59	10.6
Digital Marketing Design	62	11.2
Years of related work or study		
1-5	103	18.5
5-10	122	21.9
10-15	244	43.9
15-20	87	15.6
Frequency of using AI tools to design video ads		
Daily use	192	34.5
Use 4-6 times a week	228	41
Use 1-3 times a week	100	18
Use several times a month	36	6.5
Total	556	100

As for education, the majority of respondents held a bachelor's degree (52.7%, n = 293), followed by those with a master's degree (20.7%, n = 115). This distribution suggests that the sample was relatively well educated, which may be relevant when interpreting respondents' familiarity with emerging AI tools in creative work contexts.

With regard to background experience, 43.9% (n = 244) of respondents reported 10–15 years of related work or study experience. This combined indicator was intended to capture cumulative exposure to relevant creative fields rather than purely professional industry experience. Only 15.6% (n = 87) of respondents reported 15–20 years of related work or study experience, indicating that the sample also included a smaller proportion of respondents with comparatively longer-term exposure to relevant creative fields.

Regarding the frequency of AI tool use, 34.5% of respondents reported using AI tools for video advertising design every day, and 41.0% reported using them 4–6 times per week, indicating that such tools have become an integral part of the advertising design workflow. By contrast, only 6.5% of respondents reported using AI tools about once a month, suggesting that a small group of designers still rely on these tools only for specific tasks rather than in their day-to-day work.

V. DATA ANALYSIS AND RESULTS

Harman's single-factor test was conducted to examine potential common method variance (CMV). The first unrotated factor accounted for 31.49% of the total variance, which was below the 50% threshold, indicating that CMV was not a serious concern. In addition, the indicator-level variance inflation factor (VIF) values in the outer measurement model ranged from 1.828 to 2.486, all below the recommended threshold of 3.3, indicating that no serious multicollinearity existed among the measurement items [51]. Taken together, these results indicate that common method bias was unlikely to threaten the interpretation of the model results.

A. MEASUREMENT MODEL

The adequacy of the measurement model was examined using the Standardized Root Mean Square Residual (SRMR) and the Normed Fit Index (NFI) (Table 4). The resulting SRMR of 0.040 was well within the acceptable range (below the 0.08 criterion) [52]. Furthermore, the NFI value of 0.901 surpassed the conventional 0.90 cut-off. Collectively, these indices demonstrate a strong fit between the data and the model, validating the model's soundness.

TABLE 4. Model fit.

Fit Index	Computed Values	Threshold Reference
SRMR	0.040	[52]
NFI	0.901	[53]

Partial least squares structural equation modeling (PLS-SEM) was used to evaluate the reliability and validity of the measurement model. The ability of each indicator to represent its latent construct is assessed by its factor loading. The higher the loading, the stronger the explanatory power. As shown in Table 5, all factor loadings exceeded 0.7, ranging from 0.815 to 0.882, confirming that the measurement model has sufficient reliability and validity. Specifically, all indicators for aesthetics, authenticity, creativity, perceived trust, perceived usefulness, and behavioral intention showed high factor loadings.

TABLE 5. Factor loadings and cross-loadings.

	Ae	Au	BI	Cr	PT	PU
Ae1	0.882	0.313	0.341	0.325	0.276	0.273
Ae2	0.855	0.243	0.274	0.290	0.311	0.227
Ae3	0.866	0.271	0.303	0.270	0.298	0.256
Ae4	0.852	0.277	0.289	0.284	0.287	0.205
Au1	0.267	0.857	0.265	0.292	0.303	0.238
Au2	0.270	0.815	0.249	0.284	0.252	0.231
Au3	0.277	0.857	0.294	0.292	0.263	0.212
Au4	0.261	0.825	0.229	0.299	0.247	0.184
BI1	0.360	0.264	0.864	0.348	0.257	0.203
BI2	0.279	0.269	0.851	0.300	0.240	0.225
BI3	0.275	0.265	0.841	0.292	0.287	0.246
BI4	0.270	0.257	0.842	0.285	0.247	0.235
Cr1	0.346	0.343	0.323	0.866	0.341	0.266
Cr2	0.282	0.256	0.286	0.845	0.307	0.243
Cr3	0.272	0.289	0.305	0.853	0.295	0.303
Cr4	0.263	0.301	0.325	0.869	0.333	0.271
PT1	0.277	0.262	0.293	0.329	0.853	0.301
PT2	0.294	0.248	0.259	0.321	0.852	0.281
PT3	0.285	0.287	0.269	0.297	0.834	0.270
PT4	0.293	0.285	0.205	0.312	0.851	0.284
PU1	0.272	0.258	0.221	0.250	0.280	0.848
PU2	0.233	0.190	0.222	0.274	0.287	0.839
PU3	0.211	0.191	0.212	0.254	0.276	0.839
PU4	0.231	0.235	0.247	0.292	0.292	0.862

To establish the measurement model's psychometric properties, reliability and validity were evaluated (Table 6). Internal consistency was supported, as all Cronbach's alpha and Composite Reliability (CR) values exceeded the 0.70 threshold. Convergent validity was also affirmed, with all Average Variance Extracted (AVE) values surpassing the 0.50 cut-off [51]. Collectively, these findings validate that the measurement model is stable.

TABLE 6. Indicators of the reliability and validity of the concept.

	CA	CR (rho_a)	CR (rho_c)	AVE
Ae	0.886	0.889	0.922	0.746
Au	0.860	0.864	0.905	0.703
BI	0.872	0.874	0.912	0.722
Cr	0.881	0.882	0.918	0.737
PT	0.869	0.870	0.911	0.718
PU	0.869	0.870	0.910	0.718

Discriminant validity was assessed using the Fornell–Larcker criterion. As shown in Table 7, the square roots of AVE for all constructs were greater than their correlations with other constructs, confirming satisfactory discriminant validity [54].

TABLE 7. Discriminant validity (fornell-larcker criterion).

	Ae	Au	BI	Cr	PT	PU
Ae	0.864					
Au	0.320	0.839				
BI	0.351	0.310	0.850			
Cr	0.339	0.348	0.362	0.858		
PT	0.338	0.319	0.304	0.372	0.847	
PU	0.279	0.259	0.267	0.316	0.335	0.847

To further assess discriminant validity, this study employed the Heterotrait–Monotrait ratio (HTMT) criterion. As shown in Table 8, all HTMT values were below the conservative threshold of 0.85, providing additional evidence for satisfactory discriminant validity. Together with the Fornell–Larcker results, these findings support the discriminant validity of the measurement model.

Table 9 reports the R^2, adjusted R^2, and Q^2 values of the structural model's endogenous constructs. R^2 reflects the explanatory power of independent variables on each latent variable: perceived trust ($R^2 = 0.242$) and behavioral intention ($R^2 = 0.228$) show modest explanatory power, while the lower R^2 of perceived usefulness (0.148) still meets the acceptable threshold for exploratory behavioral research [54].

TABLE 8. Discriminant validity (HTMT values).

	Ae	Au	BI	Cr	PT	PU
Ae						
Au	0.366					
BI	0.395	0.357				
Cr	0.383	0.399	0.410			
PT	0.386	0.367	0.347	0.424		
PU	0.316	0.297	0.307	0.360	0.385	___

These modest R^2 values align with our study design: the model intentionally focuses on the net effects of three content quality dimensions (aesthetics, authenticity, creativity), and excludes classic TAM predictors (e.g., perceived ease of use) that would explain additional variance in adoption intention.

This pattern suggests that designers' usefulness evaluations in professional generative-AI advertising contexts are shaped not only by content-quality perceptions, but also by additional workflow-related, task-related, and organizational considerations beyond the scope of the current model.

The near-identical adjusted and original R^2 values confirm no substantial estimation bias and stable model performance. All positive Q^2 values (BI=0.157, PT=0.170, PU=0.103) verify the model's acceptable predictive relevance. Overall, the model demonstrates reasonable explanatory capability and predictive relevance.

TABLE 9. Values of R^2 and Q^2.

	R^2	Adjusted R^2	Q^2
BI	0.228	0.221	0.157
PT	0.242	0.237	0.170
PU	0.148	0.143	0.103

B. STRUCTURAL MODEL ANALYSIS

Before testing the structural relationships, collinearity among the predictor constructs was examined. The inner-model VIF values ranged from 1.956 to 2.691, all below the recommended threshold, indicating that multicollinearity was not a concern in the structural model. The hypothesized paths were then assessed using bootstrapping with 5,000 resamples.

The results show that aesthetics (Ae), authenticity (Au), and creativity (Cr) all have a statistically significant positive impact on perceived usefulness (PU) and perceived trust (PT), as detailed in Figure 5 and Table 10. Bootstrapping results indicate that all hypothesized paths (H1–H12) reached statistical significance at the $p < 0.05$ level.

Notably, while all paths are statistically significant, the effect sizes vary in magnitude: creativity (Cr) exerts the strongest effect on both PU ($\beta = 0.214$, $T = 4.316$,

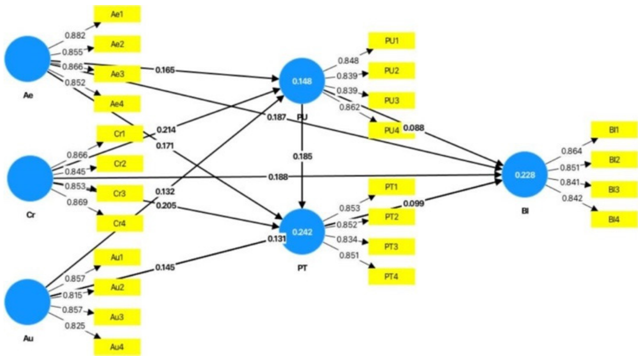

FIGURE 5. Results of PLS structural model.

$p < 0.001$) and PT ($\beta = 0.205$, T = 4.25, $p < 0.001$); by contrast, the direct effects of PU and PT on behavioral intention (BI), though statistically significant, are relatively weak in magnitude ($\beta = 0.088$, $p = 0.034$; $\beta = 0.099$, $p = 0.031$, respectively). Overall, the results support the proposed directional relationships and highlight the relevance of content-quality perceptions, perceived usefulness, and perceived trust in shaping designers' adoption intention.

Although the observed f^2 values for the structural paths range from 0.008 to 0.044, indicating mostly small effects, this pattern is consistent with prior behavioral intention research, in which multiple predictors jointly contribute to users' decision-making rather than exerting strong independent effects. According to Cohen's guidelines, f^2 values of 0.02, 0.15, and 0.35 indicate small, medium, and large effects, respectively [55]. In the present study, the relatively small f^2 values suggest that each predictor makes a modest but meaningful contribution to the endogenous constructs. These findings are reasonable in the context of designers' adoption of AI-generated video advertising tools, where perceived usefulness, perceived trust, and behavioral intention are shaped by the combined influence of multiple content-related and psychological factors rather than by any single dominant determinant.

C. MEDIATION ANALYSIS

We carried out 5,000 resamplings of bootstrapping to test the indirect effect. According to the intermediary results in Table 11, external variables will indirectly affect behavioral intent (BI) and perceived trust. Among them, creativity (Cr) has a significant impact on BI through PU ($\beta = 0.019$, 95% CI = [0.003, 0.040]). Authenticity (Au) affects BI through PU and PT respectively ($\beta = 0.048$, 95% CI = [0.023, 0.076]; $\beta = 0.116$, 95% CI = [0.070, 0.166]). In addition, PU can also further influence BI through PT ($\beta = 0.018$, 95% CI = [0.004, 0.042]), suggesting that perceived usefulness may strengthen perceived trust, which is in turn related to designers' behavioral intention.

However, the indirect effects of aesthetics (Ae) on BI through PU and PT did not reach statistical significance ($\beta = 0.014$, 95% CI = [−0.002, 0.035]; $\beta = 0.017$, 95% CI = [−0.001, 0.040]), suggesting that aesthetics did not significantly influence behavioral intention through PU or PT, although its direct effect on BI was significant.

Overall, PU and PT play important mediating roles in the relationships between content-related determinants and BI, highlighting the need to improve both the functionality and credibility of AI video advertising tools to promote adoption among advertising designers (Table 11).

TABLE 10. Hypothesis testing and collinearity assessment results.

Hypothesis	β	SD	T Value	P value	f²	Results
H1:Ae → PU	0.165	0.051	3.242	0.001	0.027	Supported
H2:Ae → PT	0.171	0.052	3.308	<0.001	0.031	Supported
H3:Ae → BI	0.187	0.049	3.837	<0.001	0.036	Supported
H4:Au → PU	0.122	0.046	2.310	0.005	0.023	Supported
H5:Au → PT	0.135	0.049	2.230	0.009	0.017	Supported
H6:Au → BI	0.151	0.050	3.387	<0.001	0.018	Supported
H7:Cr → PU	0.214	0.050	4.316	<0.001	0.044	Supported
H8:Cr → PT	0.205	0.048	4.250	<0.001	0.044	Supported
H9:Cr →BI	0.188	0.050	3.755	<0.001	0.035	Supported
H10:PU →BI	0.088	0.048	1.827	0.034	0.008	Supported
H11:PU →PT	0.185	0.051	3.641	<0.001	0.038	Supported
H12:PT →BI	0.099	0.053	1.872	0.031	0.010	Supported

VI. DISCUSSION

Grounded in the extended Technology Acceptance Model (TAM), this study examined how three content quality dimensions of AI video-generation tools—aesthetics, authenticity, and creativity—influence advertising designers' adoption intention. The results reveal that, beyond statistical significance, these dimensions operate through distinct evaluative mechanisms. Aesthetics exerted a significant direct effect on behavioral intention, yet its indirect effects via perceived usefulness and perceived trust were nonsignificant, suggesting that visual appeal may function more as an immediate cue of professionalism than as a stable basis for usefulness or trust judgments. In contrast, authenticity and creativity showed more robust indirect effects through both mediators. This pattern suggests that designers evaluate generative AI tools not merely on novelty or visual appeal, but primarily on whether outputs are trustworthy, professionally viable, and meaningfully integrable into creative workflows.

A. THEORETICAL IMPLICATIONS

First, this study refines TAM constructs for generative AI contexts. For professional users facing output uncertainty, perceived trust becomes especially important because

TABLE 11. Indirect and mediating effects.

Path	β	S.E.	T	P	95% CI Lower	95% CI Upper	Results
Cr →PU →PT	0.040	0.014	2.862	0.002	0.020	0.067	Supported
Au →PU → PT	0.024	0.012	2.076	0.019	0.008	0.048	Supported
PU →PT → BI	0.018	0.011	1.651	0.049	0.004	0.042	Supported
Ae →PU →BI	0.014	0.010	1.49	0.068	-0.002	0.035	Unsupported
Ae →PT → BI	0.017	0.011	1.591	0.056	-0.001	0.040	Unsupported
Cr →PU →BI	0.019	0.011	1.689	0.046	0.003	0.040	Supported
Ae →PU →PT	0.030	0.013	2.310	0.010	0.013	0.057	Supported
Cr →PT →BI	0.020	0.012	1.646	0.050	0.004	0.045	Supported
Au →PU →BI	0.048	0.014	3.486	<0.001	0.023	0.076	Supported
Au →PT →BI	0.116	0.024	4.826	<0.001	0.070	0.166	Supported

adoption depends not only on whether the system is usable, but also on whether AI consistently produces reliable, credible, and professionally appropriate outputs. This mechanism becomes especially important when AI-generated outputs exhibit human-like authenticity, as such outputs may strengthen perceived reliability. By empirically incorporating perceived trust into an extended TAM and testing its mediating role in generative AI-supported advertising design, this research offers a contextually grounded extension of TAM suitable for high-uncertainty, high-responsibility creative domains [20].

Second, the study shifts the adoption lens from a tool-function view to a content-output perspective. As generative AI increasingly produces the core visual and narrative elements of advertising, users' evaluations of technological value increasingly turn to the quality of the generated content itself. In this study, content quality is conceptualized through three dimensions: aesthetics, authenticity, and creativity. Aesthetics refers to whether AI-generated outputs meet professional aesthetic standards; authenticity concerns visual realism, alignment with brand tone, and narrative or character consistency; and creativity reflects the extent to which the outputs provide novel yet professionally appropriate expressions. On this basis, the study identifies a mechanism through which content quality is associated with adoption intention via perceived usefulness and perceived trust. This content-centered logic suggests that, in professional creative uses of generative AI, content quality serves as an important evaluative basis for adoption through perceived usefulness and perceived trust. The present study intentionally focuses on content-quality perceptions as a bounded explanatory pathway for designers' adoption intention, rather than attempting to incorporate all possible functional, economic, and risk-related predictors of generative-AI tool use.

Importantly, the dimensions are not uniform in their effects. Authenticity and creativity carry greater weight in shaping trust and usefulness perceptions, while aesthetics functions more as a direct, surface-level influencer. This heterogeneity highlights differentiated psychological mechanisms: aesthetic appeal alone appears insufficient to explain adoption intention through functional or trust pathways, whereas credible, contextually appropriate, and creatively valuable outputs appear more consistently associated with acceptance [56]. In generative-AI advertising, authenticity is particularly fragile—deviations risk expectancy violations and deepfake-like perceptions, elevating its role in mitigating professional and communicative risks [57].

Third, this work complements audience-centric AI advertising research by foregrounding the content creator perspective. Professional requirements for authenticity, creativity, and aesthetics serve as a front-end filter: substandard outputs are typically rejected during production, preventing flawed content from reaching audiences. The findings indicate that authenticity- and creativity-related judgments exert stronger indirect influence on adoption via usefulness and trust than aesthetics. This creator-side screening mechanism offers a necessary upstream complement to downstream consumer studies, laying theoretical groundwork for an integrated "design-to-communication" perspective on AI-generated advertising effects.

Overall, by integrating perceived trust and multidimensional content quality, this study proposes a content-centered adoption framework tailored to generative AI's unique characteristics in professional creative contexts marked by output uncertainty, evaluative judgment, and reputational risk. Adoption intention appears to be associated not only with perceived usefulness but also with whether generated outputs are deemed credible, professionally suitable, and creatively valuable. This framework refines TAM for high-uncertainty creative applications and provides theoretical guidance for AI adoption research in advertising and related industries.

B. PRACTICAL IMPLICATIONS

At the practical level, this study addresses the question of how AI-based video generation tools are adopted within the creative processes of advertising. The findings indicate that advertising designers' adoption decisions are not driven merely by "technological novelty" or "cost efficiency." Instead, adoption depends on AI's comprehensive performance across three content-related criteria: its ability to provide perceptible creative value, its capacity to maintain brand image and narrative consistency, and its conformity with professional review standards and aesthetic expectations. When AI outputs are judged to deliver reliable quality within a controllable range, these tools may be more likely to move from experimental utilities to routine creative resources. This also implies that organizations that frame generative AI merely as a low-cost acceleration tool may overestimate its practical value while underestimating the professional risks associated with poor-quality outputs.

These insights have concrete implications for designing advertising production workflows. Rather than replacing the entire production chain through a "one-click output" approach, AI video-generation tools are more effectively positioned as supportive modules during the early and middle stages of the creative pipeline. For instance, during early-stage ideation and stylistic exploration, AI can rapidly generate multiple visual alternatives to support conceptual development and tone testing. However, during the middle and final refinement stages, designers' professional judgment remains essential for screening, modifying, and reconstructing AI-generated results to ensure brand authenticity, visual coherence, and the strategic reliability of the communication outcome. This human–AI division of labor—where AI expands the creative space while humans maintain quality control—enhances efficiency while preserving the professional standards of advertising production. A workflow model that treats AI as a full substitute for professional judgment may improve short-term speed, but it is less likely to support stable brand consistency or communication reliability over time.

These findings offer evidence-based guidance for creator-side human–AI collaboration and advertising workflow governance. Aligned with recent generative AI marketing research [58], our results suggest that GenAI delivers the most sustainable advertising value as a creative augmentation tool under professional human gatekeeping, not an autonomous replacement for designers. Specifically, the stable indirect effects of authenticity and creativity suggest that designers' willingness to adopt AI tools is closely related to their ability to evaluate and refine outputs according to brand alignment, credibility, and creative validity. Meanwhile, the comparatively stronger role of creativity in relation to perceived usefulness and perceived trust indicates that designers may interpret AI-generated outputs as sources of creative support, while still retaining professional judgment as the basis for evaluation and refinement. Accordingly, effective workflow governance should embed human-led quality screening across the three core content dimensions (aesthetics, authenticity, creativity) throughout the AI-assisted production pipeline, to balance AI's efficiency advantages with the professional rigor required for brand communication.

In terms of professional competency, generative AI is shifting the role of advertising designers from "direct visual creators" to "generative strategists" and "multidimensional evaluators of content quality." Designers must be able to translate brand strategies and creative intentions into

Criterion	Operational Definition	Screening Focus	Role in Selection
Visual Quality	Overall clarity, lighting consistency, image detail, and visibility of key objects	Whether the generated clip reached an acceptable professional visual standard	Clips with blurred imagery, unstable lighting, poor object visibility, or low detail were excluded
Motion Smoothness	Temporal continuity of camera movement and object motion, with minimal jitter or abrupt transitions	Whether the video appeared fluent and visually stable	Clips with obvious motion artifacts, disruptive transitions, or jitter were excluded
Stylistic Coherence	Consistency with the unified prompt, perfume-ad tone, and continuity	Whether the clip matched the intended atmosphere and style of the stimulus	Clips inconsistent with the prompt, ad style, or scene sequence were not prioritized
Advertising Appropriateness	Suitability for perfume advertising in terms of product emphasis, mood, and brand-like presentation	Whether the clip could function as part of a plausible perfume advertisement	Clips lacking product salience, mood resonance, or advertising relevance were excluded

structured prompts that embed brand elements, visual styles, and situational cues to guide the model toward producing appropriate outputs. Additionally, they must systematically evaluate large volumes of AI-generated content based on aesthetic quality, authenticity, and creative strength, distinguishing between outputs that are merely "technically feasible" and those that qualify as high-value brand assets. Correspondingly, advertising and design education should evolve from basic software-skill training to the systematic cultivation of AI literacy, critical evaluation ability, human–AI co-creation methods, and heightened sensitivity to ethical issues such as authenticity, attribution, and copyright.

From an industry-development perspective, AI significantly lowers the marginal cost of content generation and iteration, making "multi-version testing, multi-touchpoint distribution, and continuous optimization" a feasible

Variable	Measurement items	Source of adaptation
Authenticity	The AI-generated video ad makes a genuine impression. The AI-generated video ad looks real. The AI-generated video ad appears authentic. The AI-generated video ad gives an impression of being natural.	[59]
Aesthetics	The AI-generated ad images are highly visually appealing. The AI-generated video ad is pleasant to look at. The AI-generated video ads are clear, intuitive, and visually coherent. The overall style of the ad is coordinated and consistent.	[60]
Creativity	The AI-generated ads are creative. The AI-generated ads are innovative. The AI-generated ads show originality. The AI-generated ads are imaginative.	[61]
Perceived Usefulness	Using AI video tools will improve my design performance. Using AI video tools will help my work. Using AI video tools will improve my work efficiency. I think AI video tools are useful for my design work.	[14]
Perceived Trust	If I think the content generated by AI tools is trustworthy, I will use AI video tools to make video ads. If I think the information provided by AI tools is reliable, I will use AI video tools to make video advertisements. If I think AI tools can meet my expectations, I will continue to use AI video tools to generate video ads. If I think the AI tool is safe, I will use the AI video tool to make video ads.	[62]
Behavioral Intention	I will use AI video advertising tools regularly in the future. I will strongly recommend others to use AI video advertising tools. I will use AI video advertising tools frequently in the future. I plan to use AI tools to find solutions when I face video advertising design challenges.	[63]

strategic model. This shift is fundamentally reshaping the production and dissemination logic of video advertising. The results of this study suggest that, compared with traditional linear workflows that rely on producing a single "blockbuster" asset, AI-assisted workflows—centered on content quality and balancing authenticity, creativity, and aesthetic excellence—are more likely to serve as sustainable adoption pathways in the future advertising industry. Organizations that proactively adjust their workflows, task structures, and competency-development systems will be better positioned to convert generative AI into a competitive creative advantage.

In summary, the practical significance of this study lies in emphasizing that the advertising industry must actively respond to both the opportunities and risks posed by AI by adapting work modes, evaluation standards, and competency frameworks. Keeping content quality central and treating human–AI collaboration as a guiding principle may support the more sustainable adoption and integration of AI-driven video-generation tools into advertising practice. These findings also imply that organizational adoption of generative AI requires not only workflow redesign but also governance mechanisms capable of managing authenticity risk, accountability, and the strategic consequences of unreliable AI-generated outputs.

VII. LIMITATIONS AND FUTURE RESEARCH DIRECTIONS

This study focused on Chinese advertising designers and combined a stimulus-based scenario with a questionnaire survey to explore how content quality affects users' intention to use AI video-generation tools. Although this study offers theoretical and practical implications, there are still some limitations. First, the sample was mainly drawn from advertising designers and related creative practitioners in mainland China. Their professional training background, industry ecology, and technological usage environment have certain characteristics. This situation helps to gain an in-depth understanding of the application scenarios of generative artificial intelligence in the local advertising industry, but it also limits the external generalizability of the research conclusions to a certain extent. Differences in cultural expectations regarding advertising authenticity, industry structure, and local AI tool ecosystems may all influence designers' judgments of aesthetics, authenticity, creativity, trust, and adoption intention.

Second, the video materials used were generated from standardized prompts, which may not fully reflect the complexity of real advertising workflows. Third, this study only involves video forms and does not cover other media types, such as images or interactive advertisements, which may affect the evaluation of the relative importance of different content attributes. Artificial intelligence tools are undergoing rapid iteration. This study only presents designers' cognition and adoption intention at a specific point in time. In addition, the present content-centered framework explains only part of perceived usefulness in professional advertising-design contexts, suggesting that designers' usefulness evaluations may also be shaped by additional functional, workflow-related, and organizational factors beyond the scope of the current model. Some perceived-trust items in this study also partly incorporated future-use language, which may have created limited conceptual proximity to behavioral intention, even though the discriminant-validity statistics were acceptable.

Future research can further explore several directions. Methodologically, future research could adopt more strictly differentiated trust measures to reduce potential semantic overlap between trust evaluation and usage intention. In addition, future studies could extend the present content-centered framework by incorporating additional functional, workflow-related, and organizational predictors, such as perceived cost, ease of use, copyright concerns, job insecurity, controllability, and other relevant contextual factors. First, multi-country replication studies can be conducted to test whether the model of this study remains stable across different cultural contexts, industry structures, and AI tool ecosystems. Such studies would help further clarify the generalizability of the findings of this study. Second, through longitudinal or scenario-based research designs, future studies are expected to get closer to real workflows and reveal how designers gradually learn, adapt, and form stable interaction patterns with artificial intelligence tools in practice.

At the theoretical level, concepts such as perceived risk, sense of control, or task–technology fit (TTF) can also be introduced to deepen the understanding of adoption motivations and barriers. For example, different stages of the creative process, such as concept generation, style exploration, and post-production optimization, have different requirements for technical support. Future research can also systematically compare the applicability of artificial intelligence tools in these stages and the evolution of their mechanisms, so as to promote the deep integration of artificial intelligence and the advertising creative industry.

VIII. CONCLUSION

From the perspective of advertising designers, this study examined how the perceived content quality of AI-generated video advertisements—aesthetics, authenticity, and creativity—is associated with adoption intention through perceived usefulness and perceived trust. The findings show that all three attributes are related to adoption intention, but through different paths: authenticity and creativity show more stable indirect associations through usefulness and trust, whereas aesthetics shows a stronger direct association. Creativity has the strongest associations with usefulness and trust, suggesting that designers may increasingly view AI as not only a technical aid but also a creative partner. In addition, in generative-AI advertising contexts marked by output uncertainty and professional evaluation, perceived trust appears to be a particularly important mechanism associated with adoption intention.

Theoretically, this study refines TAM in the context of generative AI by showing that content-quality perceptions are associated with designers' adoption intention through usefulness and trust. It also highlights the importance of evaluating AI-generated outputs in professional creative settings.

Practically, the findings suggest that the adoption of AI video-generation tools in advertising is more closely associated with whether outputs are perceived as useful, trustworthy, and professionally suitable than with efficiency alone. This implies that the sustainable integration of generative AI into advertising practice requires not only technical improvement but also continued attention to content quality and human–AI collaboration.

APPENDIX A
EXPERT SCREENING CRITERIA FOR REPRESENTATIVE VIDEO CLIPS

See Page 13.

APPENDIX B
QUESTIONNAIRE FOR VARIABLE ITEMS AND REFERENCE

See Page 14.

DECLARATIONS

Author Contributions: Conceptualization: Ming Liu and Jinho Yim; methodology: Ming Liu; software: Ming Liu; validation: Ming Liu and Jinho Yim; formal analysis: Ming Liu; investigation: Ming Liu; data curation: Jinho Yim; writing—original draft preparation: Ming Liu; writing—review and editing: Ming Liu; visualization: Ming Liu; supervision: Jinho Yim. All authors have read and agreed to the published version of the manuscript.

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INSTITUTIONAL REVIEW BOARD STATEMENT

All subjects gave their informed consent for inclusion before they participated in the study. Ethics approval is not required for this type of study. The study was conducted following the local legislation: https://www.law.go.kr/LSW//lsLinkCommonInfo.do?lspttninfSeq=75929&chrClsCd=010202. (accessed on 10 January 2025)

INFORMED CONSENT STATEMENT

Informed consent was obtained from all subjects involved in the study.

DATA AVAILABILITY STATEMENT

All data generated or analyzed during this study are included in this article. The raw data are available from the corresponding author upon reasonable request.

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